PORTFOLIO PROTOTYPE

Text-to-SQL agent

Ask a database a question in plain English, get read-only SQL you can check, and see exactly what the model was given.

SQLite, DuckDB and PostgreSQL local Ollama or hosted Gemini

Evaluated on Spider, BIRD and a 1,222-question large-schema suite

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Accurate enough to demo, measured enough to trust the numbers

Where the agent stands on 5 November 2026, and what the October work established

What it is

A question becomes **one read-only SELECT**, checked by the application and run under each engine's own read-only mode. Answers show the rows, the SQL and the schema text sent. Result rows never reach the model.

How well it answers

64.6 % on Spider dev and **35.2 %** on BIRD dev with evidence, with a local 9B model and 95 % intervals on every figure.

What October established

On 21- to 65-table databases, sending the top 6 tables scored **+3.6 points** over full DDL plus hints, with **69 % fewer prompt tokens**. Five later ideas were measured; none was a measurable accuracy gain.

What is next

Gemini under the same contract, hint coverage beyond the 20-value ceiling, and recovering the **128 questions** whose needed table is not sent.

Useful answers need checkable SQL and a database that cannot change

The problem, and exactly what is sent to the model

WHAT THE MODEL SEES

1	**Schema metadata**	Table and column names, types, keys, CREATE TABLE DDL	**sent**
1	**Value hints**	Up to 6 frequent values of text columns with at most 20 distinct values; a cut list says so	**sent, bounded**
2	**Question value matching**	Stored values matching words in the question, from any text column	**built, off**
2	**Result rows**	The answer's rows	**never sent**

1 The model sees structure

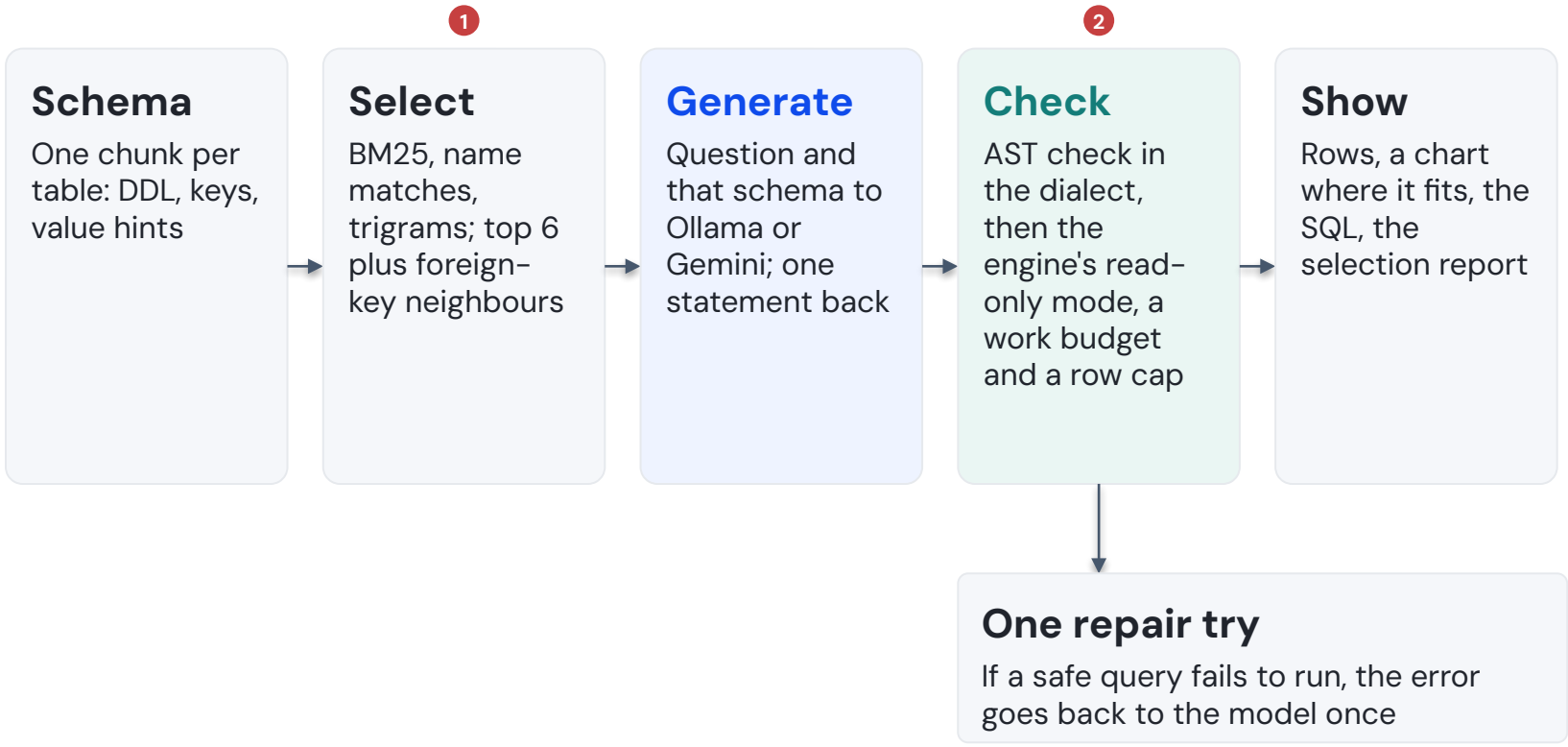
Names, types, keys and DDL, plus a few category values so it can spell a filter. That is enough to write SQL without reading the data.

2 Metadata, not data

Rows stay local. The one feature that would send stored values matching the question was measured and kept off.

The model proposes the SQL; the application decides whether it runs

One question, five steps, from schema to answer



1 Selection is the only model-facing choice

Which tables the model sees is decided before generation, and the app reports which were picked and why.

2 Execution authority stays in Python

Unsafe SQL is **refused before any database call**; a failed safe query gets one repair attempt.

Two independent read-only guarantees on every engine

What the application checks, what each engine enforces, and how it is tested

Engine	Application check	Engine guarantee
SQLite	one SELECT or WITH, parsed in this dialect; internals blocked	read-only URI, query_only, an authorizer denying writes, attach and pragmas
DuckDB	one SELECT or WITH, parsed in this dialect; internals blocked	opened read_only with external access off
PostgreSQL	one SELECT or WITH, parsed in this dialect; internals blocked	least-privilege role in a read-only transaction; refuses to connect as a superuser

45 / 0
conformance tests passed / skipped, 15 per engine

1,870
tests passed against live PostgreSQL 16, 4 Nov

6 / 6
data-change requests refused on the safety suite

1 Each guarantee holds alone

The same conformance suite runs on every engine with **no skips allowed**; neither the parser nor the engine is trusted by itself.

2 Refusing is measured too

The model declined every write request. Internals requests came back as unanswerable rather than refused, and the gate still blocked them.

Scope: safety accuracy 73.3 % [53.3, 93.3] on 15 cases.

Every reported number comes from a citable, paired run

The contract behind the results, and the suites it runs on

Suite	Questions
Spider 1.0 dev ≤ 14 tables per database	1,034
BIRD dev with and without evidence hints	1,534
[2] **bird_train_large** 4 databases of 21 to 65 tables	1,222
demo the 3 demo databases	12
safety refusals and unanswerables	15

[1]

Typed result comparison

EX compares result sets by type, not SQL text

95 % bootstrap intervals

over cases, on every rate

A manifest per run

commit, suite and database hashes, prompt hash, model, settings

Paired comparisons

two runs differing in one setting, same questions

[1] Comparisons, not single scores

Each change is judged **paired on the same questions** against a named baseline, so small effects are not read from separate intervals.

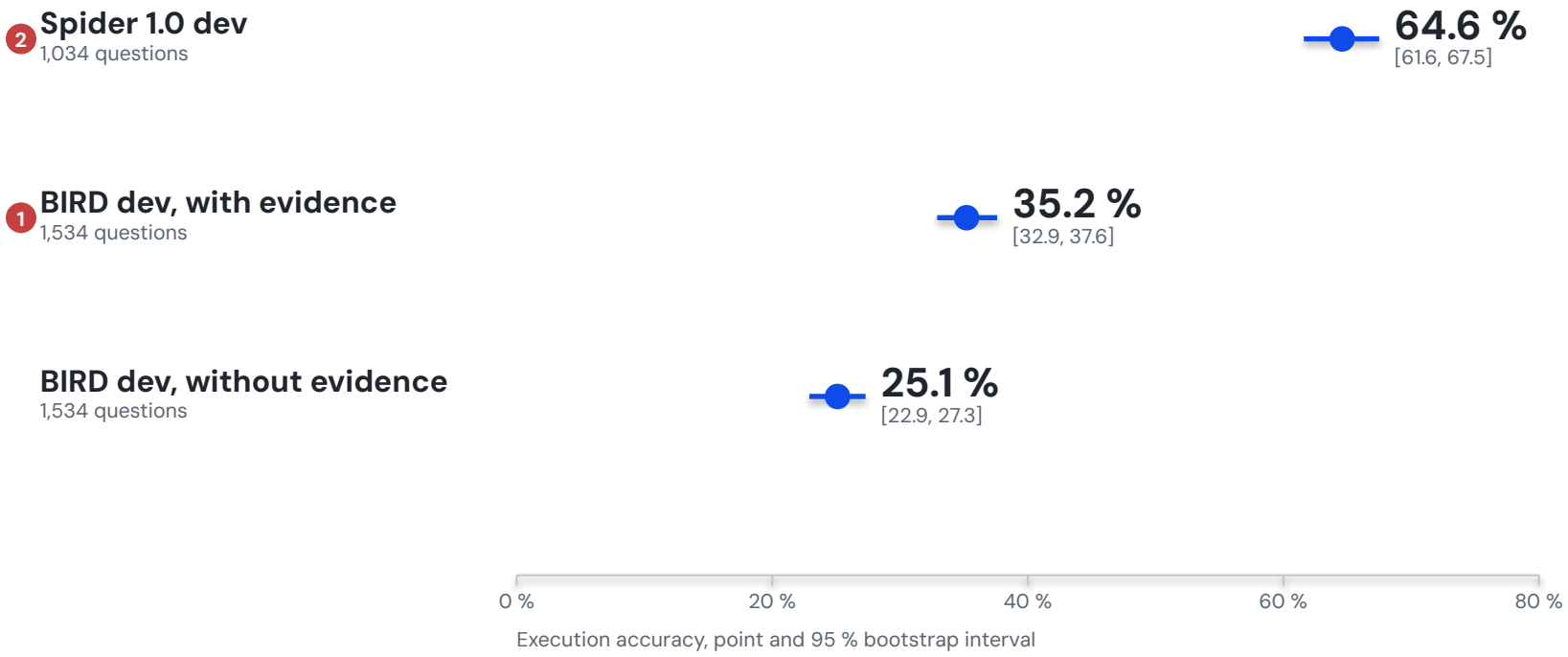
[2] A suite built for the question

Spider and BIRD dev have too few tables to show whether choosing tables helps, so a large-schema suite was built from BIRD train.

Note: EX here is this contract's typed comparison, not each benchmark's scorer.

A local 9B model answers about two in three Spider questions

Execution accuracy on full dev sets, Ollama qwen3.5:9b, thinking off, 10 Oct 2026



1 Evidence hints are worth 10 points

On the same BIRD questions, the benchmark's hints add **10.1 points [8.1, 12.1]**.

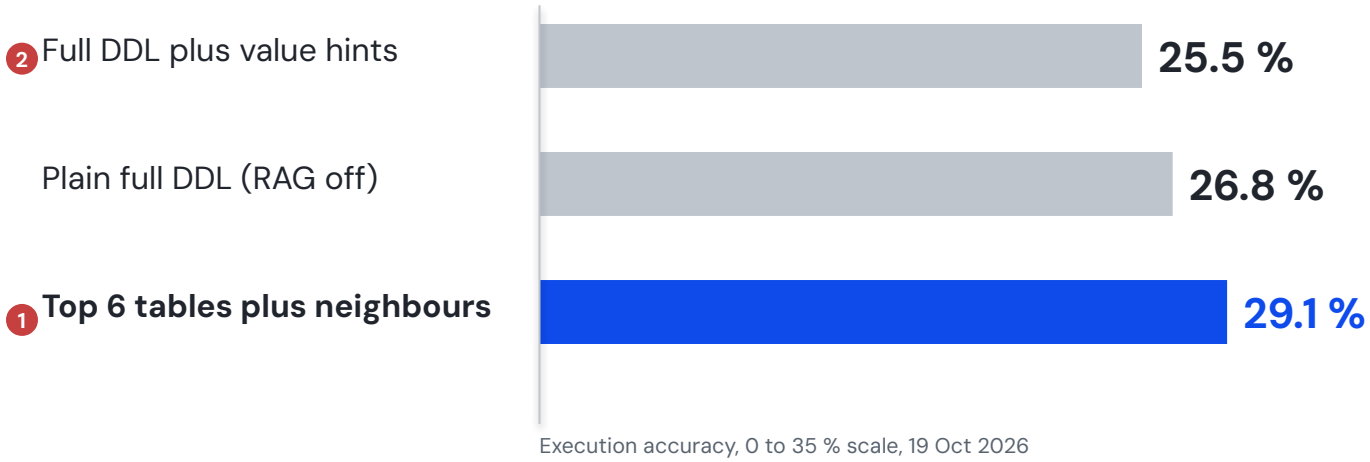
2 Thinking off was the largest fix

With thinking on, long thoughts used the output cap and no SQL came back. Off: **+8.5 points [5.8, 11.3]** on Spider, paired.

Note: not comparable with leaderboard figures, which use each benchmark's scorer.

On large schemas, choosing tables beats sending them all

Execution accuracy on bird_train_large: 1,222 BIRD train questions, 4 databases of 21 to 65 tables



Execution accuracy, 0 to 35 % scale, 19 Oct 2026

+3.6 points
[+1.4, +5.8] over full DDL plus hints, paired

69 % fewer
prompt tokens than full DDL plus hints

3
About 2 s slower
per question: each prompt's schema differs, so Ollama cannot reuse its cache

1 **Selection pays where there is choice**

The 65- and 34-table databases carry the gain, **about 5 to 6 points each**. On Spider and BIRD dev it added nothing measurable.

2 **No measurable effect from hints here**

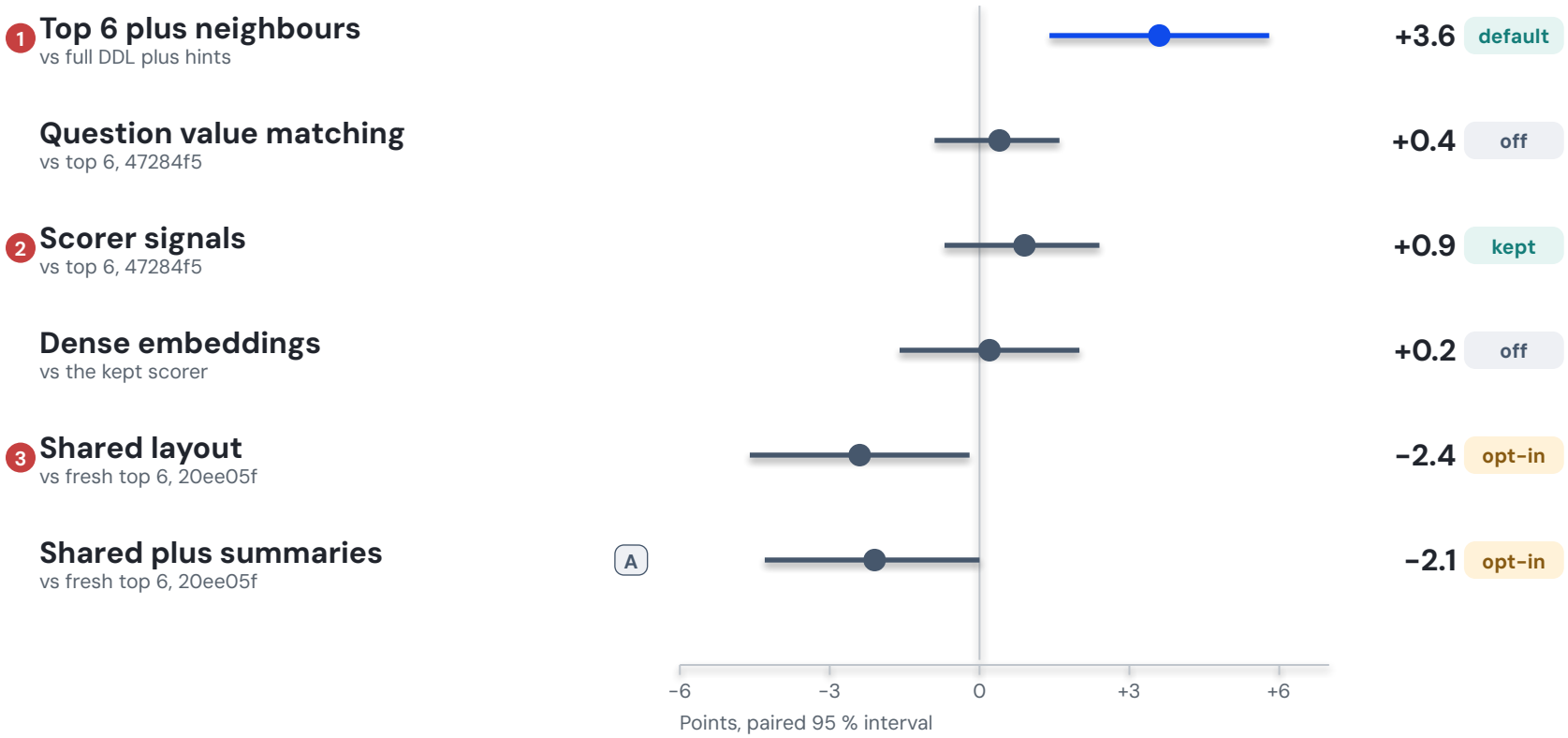
Full DDL plus hints against plain DDL: -1.3 points [-2.9, +0.2]. Half the needed literals sit in columns over the 20-value ceiling.

3 **The cost is time**

The default is slower but smaller and more accurate; the fast layouts on the next slide trade accuracy back.

Only selection delivered a measurable accuracy gain

Paired change in execution accuracy on the 1,222 questions, each against its named baseline



1 Only selection is a measured gain

Its interval is the only one **entirely above zero**.

2 Kept for retrieval, not for accuracy

The scorer change raised retrieval MRR from 0.853 to 0.891; its +0.9 on answers is inside its interval.

3 Faster prompts cost accuracy

Sending every table lets Ollama reuse its cache: 58 % faster, 2.4 points lower. Why is not settled; length, content and placement changed together.

Note A: compared across commits, so not a controlled comparison.

Sending the right tables more often did not mean more right answers

Correct answers by subgroup on bird_train_large, under the default and two full-schema layouts

Prompt layout	3 128 with a needed table not sent	The other 1,094	Seconds per question
Selected tables (default)	**8**	**359**	**10.2**
2 Every table, selected named	**23**	**315**	**4.3**
Every table, plus summaries	**21**	**320**	**4.4**

1 Dense embeddings, against the kept scorer

Questions with a needed table not sent: **128 to 74**. Correct among the original 128: **8 to 26**. Whole suite: **+0.2 points [−1.6, +2.0]**.

1 Recall is not accuracy

Embeddings fixed many missing-table questions, but changed table sets flipped about as many other answers the wrong way. Both are reported, separately.

2 The few gain, the many lose

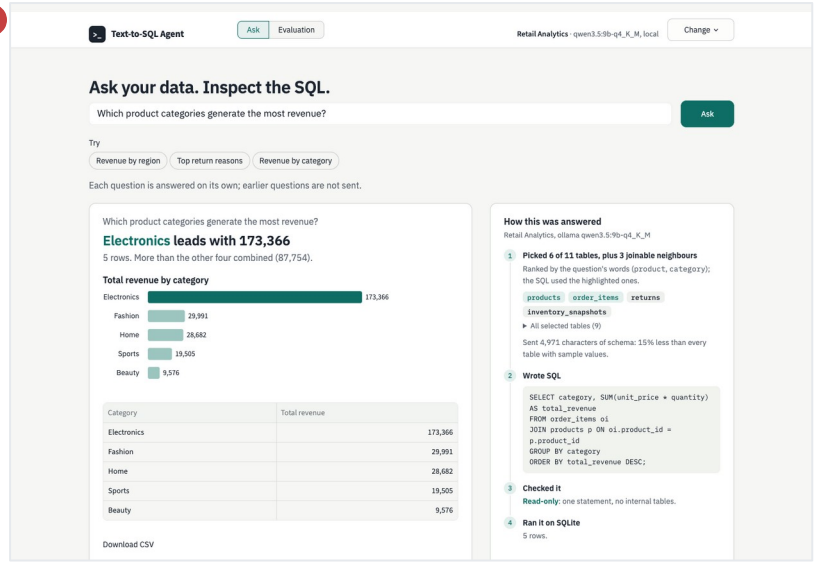
Sending every table helps the 128 and costs the 1,094 more. **That is what these layouts did**, not a limit on better retrieval.

3 Next test

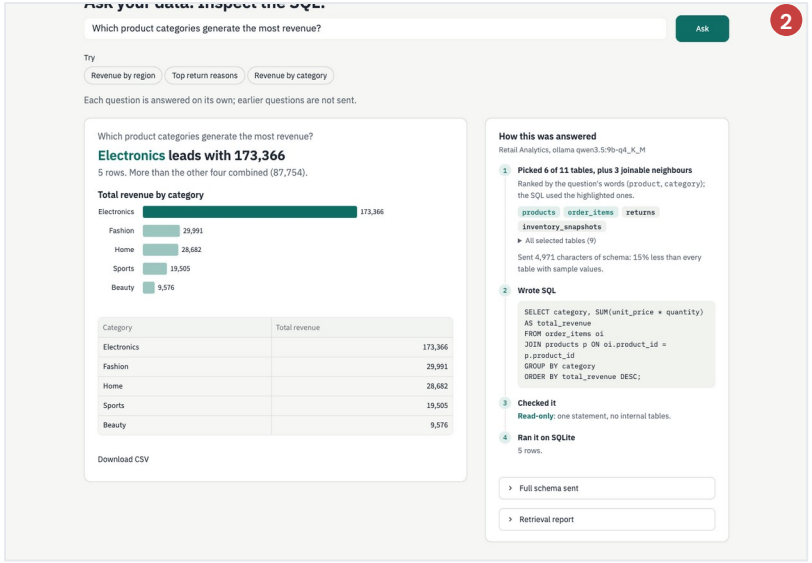
Recover the missing tables for the 128 without changing what the other 1,094 receive.

The answer comes first; the evidence is one glance away

Redesigned Streamlit interface, real captures from a local model on the demo database



Answer: headline from the rows, chart, table



How this was answered: tables and why, SQL, the check, the run

1 Answers state what the rows say

A headline from the result, a bar chart when the result is one category and one number, and the table.

2 Evidence is recorded, not rebuilt

Tables picked and why, the schema size sent, the SQL and the read-only check are **kept from the moment of answering**.

NEXT STEPS

What the results do not yet show, in priority order

Limits to keep beside the numbers, and the tests that would address them

1 Measure Gemini under the same contract

Supported but unmeasured since May. Its thinking shares the 512-token output cap; set a thinking budget and record it in the run identity once a key is available.

2 Recover the 128 missing-table questions

Find their tables without lengthening the prompt for the other 1,094.

3 Widen hint coverage

Half the needed literals sit in columns over the 20-value ceiling. Listing more values needs a privacy decision first.

4 Test more large schemas

The selection gain comes from two of four BIRD train databases; other schemas are untested.

Scope: all v2 accuracy figures use one local model, qwen3.5:9b-q4_K_M via Ollama; EX is this project's typed comparison, not a leaderboard score.

THANK YOU

Grounded, visible, read-only and measured, including the ideas that did not work.

Live demo aipa-text-to-sql-agent.streamlit.app

Code, runs and decision log github.com/tuannm3812/text-to-sql-agent

Began as a UTS group project in May 2026 with Huy Duc Vu, Ngoc Bao Tran Nguyen, Manh Tuan Nguyen and Nguyen Nghia Doan Dang; refined since by Manh Tuan Nguyen.